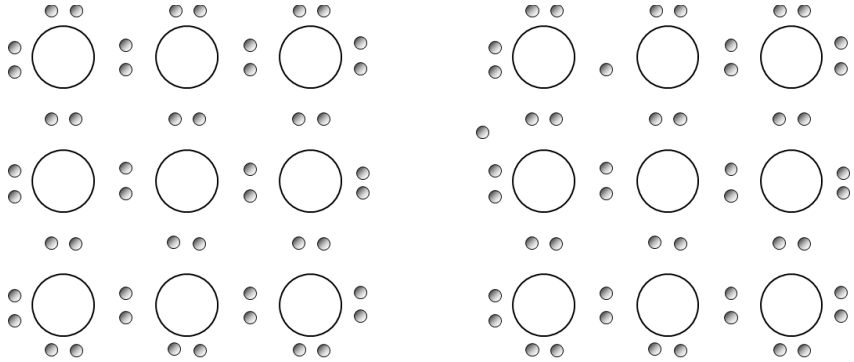
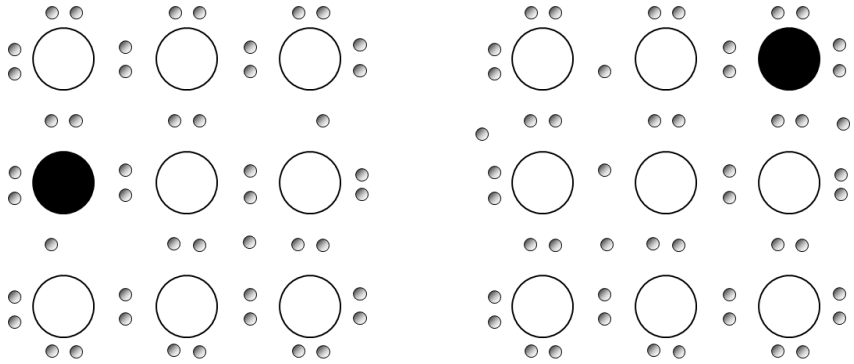
EEEN20070 tutorial questions
Chapter 1: Semiconductor Physics

1. Sketch a simplified 2D Si lattice consisting of 9 Si atoms and identify their valence electrons. Make the sketch for T=0K. How would this sketch change is T=300K?



0K T>0K

2. Repeat the sketch of question 2 (T>0K) but for:
   i) p-type Si
   ii) n-type Si



p-Si n-Si

p-type doping: boron (B) introduces holes
n-type doping: phosphor (P) introduces electrons

3. Holes in semiconductors are – choose:
   ~~a. Voids in the lattice where the atom is missing~~
   b. A positively charged carrier that can freely roam around the lattice at T = 300K.
   ~~c. A proton in the nucleus of the atoms building up the lattice.~~

4. Give the law of mass action formulae (know by heart) and the expression for charge neutrality (know by heart) and derive the expressions for the electron and hole concentration in p-type Si.

$$n \times p = n_i^2$$

$$e\left(-n + p + N_D^+ - N_A^-\right) = 0$$

$$-n + p + N_D^+ - N_A^- = 0$$

$N_A >> N_D \approx 0$ p-doped Si

$$-n + p - N_A^- \approx 0$$

$$-\frac{n_i^2}{p} + p - N_A^- \approx 0$$

$$p^2 - N_A^- p - n_i^2 \approx 0$$

$$p = \frac{N_A^- \pm \sqrt{N_A^{-2} + 4n_i^2}}{2}$$ General solution of quadratic equation

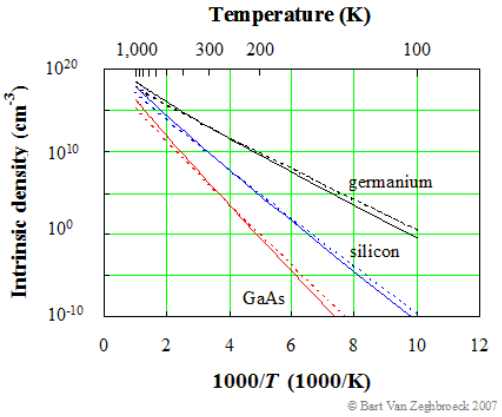
$$p = \frac{N_A^- + \sqrt{N_A^{-2} + 4n_i^2}}{2}$$ Density cannot be negative

$$p = \frac{N_A^- + N_A^- \sqrt{1 + 4\frac{n_i^2}{N_A^{-2}}}}{2}$$

$N_A^- >> n_i$ at room temperature and normal doping densities

$10^{15} cm^{-3} < N_A^- < 10^{20} cm^{-3}$

at higher temperatures n_i increases: $n_i = \sqrt{N_C N_V} \exp\left(\frac{-E_G}{2kT}\right)$ (derived later)



$$p \approx \frac{N_A^- + N_A^- \sqrt{1}}{2}$$

$$p \approx N_A^-$$

Use law of mass action to get:

$$n = \frac{n_i^2}{p} = \frac{n_i^2}{N_A^-}$$

5. Calculate the concentration of holes and electrons (at room temperature) for:
   In order to simplify your calculations, you can use appropriate approximations.

   i) Intrinsic silicon
      Look up in table of constants: $n_i = 1.45 \times 10^{10}$ cm^{-3} = p = n
   ii) Boron doped silicon: [B]=10^{19} cm^{-3}

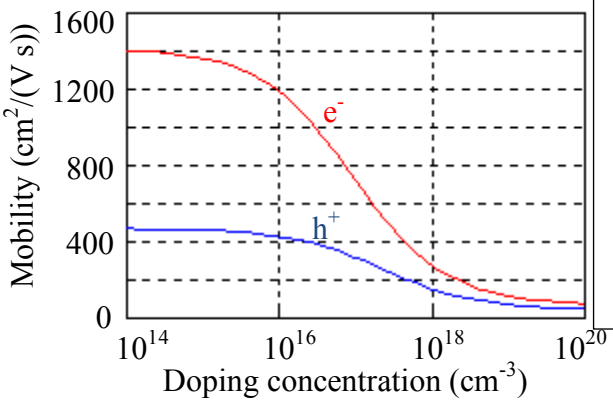
      $p_p = 10^{19}$ cm^{-3}, $n_p = \frac{n_i^2}{N_A^-} = \frac{(1.45 \times 10^{10})^2}{10^{19}} = 21$ cm^{-3}
   iii) Phosphorous doped silicon: [P]=10^{24} m^{-3}

$n_n = 10^{18}$ cm^{-3}, $p_n = \frac{n_i^2}{N_D^+} = \frac{(1.45\times10^{10})^2}{10^{18}} = 210$ cm^{-3}

6. The mobility of electrons and holes in Si is different and depends on the doping concentration. Give the reason why the mobility depends on doping concentration.

$v = \frac{q\tau}{m}E \rightarrow v = \mu E \rightarrow \mu = \frac{q\tau}{m}$ One of the parameters is τ, the average time between scattering events. Carriers will travel through a periodic lattice with a certain velocity determined by the interaction of the carriers with the lattice – called scattering. If changes happen to this periodicity, the interactions will change. Disrupting the lattice with foreign atoms (as with doping atoms) will increase the scattering events and decrease the average lifetime τ between these events.

7. Figure 2 gives the plot of the carrier mobility as a function of doping concentration.



Note that the mobility of holes is approximately half that of electrons in Si. Both electron and hole mobility are dependent on the doping concentration. Thus for $N_D = 10^{18}$ cm^{-3} $\mu_n \approx 300$ cm^2/(Vs) and $\mu_p \approx 150$ cm^2/(Vs).

Figure 1: The variation of the carrier mobility as a function of doping density.

Four Si resistors are given with geometry as in figure 2. The only difference between the four samples is the doping density and type, given in table 1. The doping is homogeneous. Calculate the electrical conductivity, σ_i at room temperature of the four samples and calculate their resistance. The effective mass in Si of the electrons is 0.26 and for the holes 0.36. The free electron mass is $m_0 = 5.11\ 10^{-31}$ kg.

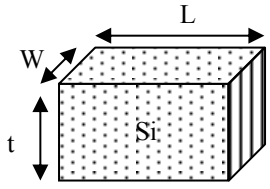


Figure 2: Geometry of the Si samples. W=1 μm, t = 500 nm L = 5 10^{-6} m.

Sample n°	1	2	3	4
N_A (cm^{-3})	0	10^{15}	10^{16}	0
N_D (cm^{-3})	0	0	0	10^{16}

Table 1: Sample number and doping.

The expression for the electrical conductivity is implicitly available in the formulae list:

$$J_n(x) = e\mu_n n(x)E(x) + eD_n\frac{dn(x)}{dx} \quad \text{thus} \quad \sigma_n = e\mu_n n$$

$$J_p(x) = e\mu_p p(x)E(x) - eD_p\frac{dp(x)}{dx} \quad \sigma_p = e\mu_p p$$

Sample 1 is intrinsic: lowest carrier concentration, lowest conductivity

Sample 2 is p-type: lowest doping concentration, second lowest conductivity
Sample 3 and 4 have the same doping concentration but sample 3 has holes as majority carriers with a lower mobility than sample 4 where electrons are the majority carriers thus:
$\sigma_1 < \sigma_2 < \sigma_3 < \sigma_4$
Note that the electrical conductivity is independent of dimensions of the material.

1.
$$\sigma_n = e\mu_n n_i = 1.610^{-19} C \times 1400 \frac{cm^2}{Vs} \times 1.4510^{10} cm^{-3} = 3.2510^{-6} \frac{1}{\Omega cm}$$
$$\sigma_p = e\mu_p p_i = 1.610^{-19} C \times 500 \frac{cm^2}{Vs} \times 1.4510^{10} cm^{-3} = 1.1610^{-6} \frac{1}{\Omega cm}$$
$$\sigma_{tot} = \sigma_n + \sigma_p = 4.4110^{-6} \frac{1}{\Omega cm}$$
Conductivity of intrinsic Si is very small.

2. Since the contribution to the total conductivity of the minority carriers is very small (see sample 1) we can ignore it.
$$\sigma_{tot} \approx \sigma_p = e\mu_p p = 1.610^{-19} C \times 500 \frac{cm^2}{Vs} \times 10^{15} cm^{-3} = 8 \times 10^{-2} \frac{1}{\Omega cm}$$
3. $$\sigma_{tot} \approx \sigma_p = e\mu_p p = 1.610^{-19} C \times 400 \frac{cm^2}{Vs} \times 10^{16} cm^{-3} = 0.64 \frac{1}{\Omega cm}$$
4. $$\sigma_{tot} \approx \sigma_n = e\mu_n n = 1.610^{-19} C \times 1200 \frac{cm^2}{Vs} \times 10^{16} cm^{-3} = 1.92 \frac{1}{\Omega cm}$$

Calculation of the resistance
$$R = \frac{\rho L}{tW} = \frac{L}{\sigma_{tot} tW}$$
In order to avoid making dimensional errors, convert everything into cm.
W=1 μm = 10^{-6} m = 10^{-6} (10^2 cm) = 10^{-4} cm
t = 500 nm = 500 10^{-9} m = 500 10^{-9} (10^2 cm) = 500 10^{-7} cm = 5 10^{-5} cm
L = 5 10^{-6} m = 5 10^{-4} cm
$$R = \frac{1}{\sigma_{tot}} \frac{L}{tW} = \frac{1}{\sigma_{tot}} \frac{510^{-4}}{510^{-5} \times 10^{-4}} = \frac{10^5}{\sigma_{tot}}$$
$$R_1 = \frac{10^5}{4.4110^{-6}} \approx 2\,10^{10}\,\Omega$$
$$R_2 = \frac{10^5}{810^{-2}} \approx 10^6\,\Omega$$
$$R_3 = \frac{10^5}{0.64} \approx 1.6\,10^5\,\Omega$$
$$R_3 = \frac{10^5}{1.92} \approx 5\,10^4\,\Omega$$

8. Assume that, in an n-Si layer at T = 300K, the electron concentration varies linearly from 10^{18} cm^{-3} to 7×10^{17} cm^{-3} over a distance of 0.10 cm. calculate the diffusion current density if the electron diffusion coefficient is 50 cm^2/s.

The electron diffusion current is:

$$J_n(x) = eD_n \frac{dn(x)}{dx}$$

$= 1.6\times10^{-19}$ C×50 cm^2/s (10^{18} cm^{-3} - 7×10^{17} cm^{-3})/0.1 cm = 24 A cm^{-2}

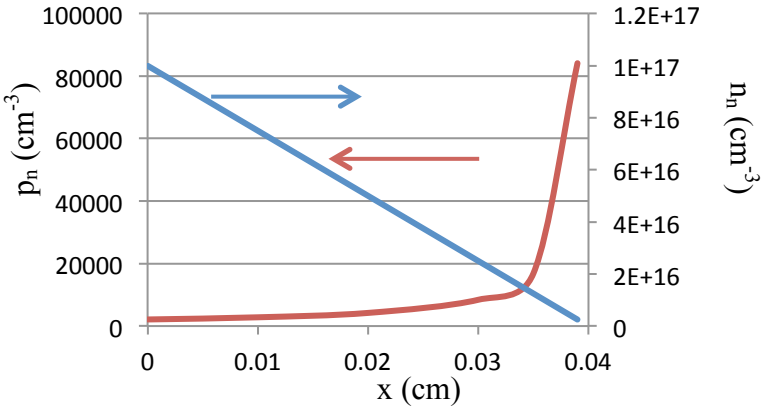
9. Assume that the doping concentration in n-type Si is not homogeneous but varies slowly. The Si sample has a length of 0.039 cm. The variation of the doping concentration is defined by the function:

$$N(x) = 10^{17}\ (\text{cm}^{-3}) - 2.5 \times 10^{18}\ (\text{cm}^{-4})\ x\ (\text{cm})$$

i) Sketch the variation of the concentrations of electrons and holes in this layer when no bias is applied for .

$n_n(x) \approx N(x)$
$p_n(x) \approx n_i^2/N(x) = n_i^2/(10^{17} - 2.5 \times 10^{18}\ x)$

Note: if the material layer is 0.04 cm long, then when N(x) → 0 then $n_n \rightarrow n_i$ $p_n \rightarrow n_i$ (if no doping then the carrier concentration becomes the intrinsic carrier concentration)



ii) Derive the expression for the internal electric field. Eliminate all material parameters so that the equation is only a function of constants and x.

Open circuit, no bias applied thus current density has to be zero:

$$J_n(x) = e\mu_n n(x)E(x) + eD_n \frac{dn(x)}{dx} = 0$$

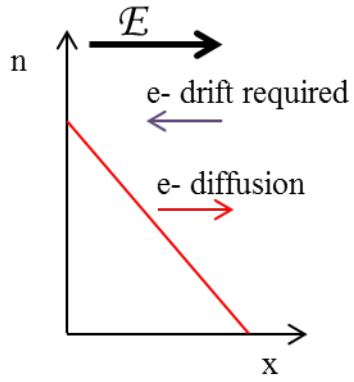
$$e\mu_n n(x)E(x) = -eD_n \frac{dn(x)}{dx}$$

$$E(x) = -\frac{D_n}{\mu_n}\frac{1}{n(x)}\frac{dn(x)}{dx}$$

$$E(x) = -\frac{kT}{q}\frac{1}{n(x)}\frac{dn(x)}{dx}$$

$$E(x) = 0.026\frac{2.5\times10^{18}}{10^{17} - 2.5\times10^{18}x} = \frac{0.065}{0.1 - 2.5x}$$

iii) Draw the direction of the internal electric field.
Start point: the gradient in electrons causes a diffusion flux of electrons from left to right. Thus for the total current to be zero an electron drift flux from right to left is needed. The electric field that is needed is in the opposite direction as the electron drift flux needed.



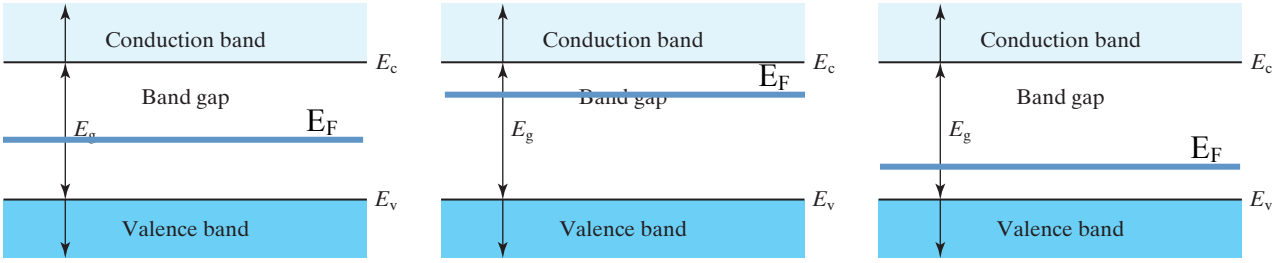
10. The probability that a state is empty is:

$$1-f(E)=1-\frac{1}{1+\exp\left(\frac{E-E_F}{kT}\right)}$$

$$0.01=1-\frac{1}{1+\exp\left(\frac{5.95-6.25}{kT}\right)}$$

thus kT = 0.06529 eV which gives T = 756 K = 483 °C. Thus the Fermi probability function is very sensitive to temperature.

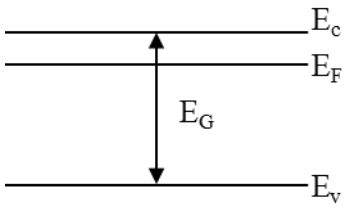
11. Give the probability of finding an electron at E = E_F in the three following situations.



(a) (b) (c)

It is ½ in all of the 3 cases. The probability of finding an electron at an energy equal to the Fermi level is ½. This is the definition of the Fermi level. It means that if the Si is doped with donor atoms, the Fermi level needs to shift up such that it can represent more electrons in the conduction band.

12. Draw the energy band diagram (E_c, E_v, E_F, E_G) of n-type Si.



13. The intrinsic level, E_i in a semiconductor is defined as the position of the Fermi level, E_F for an intrinsic material. Using this definition write the electron concentration as a function of n_i and $E_i - E_F$. Do the same for the hole concentration.

In the formulae list the concentration of electrons and holes, based on Fermi-Dirac statistics, is given

$$n = N_C \exp\left(\frac{-(E_c - E_F)}{kT}\right)$$

$$p = N_V \exp\left(\frac{-(E_F - E_v)}{kT}\right)$$

Definition of intrinsic level:

$$n_i = N_C \exp\left(\frac{-(E_c - E_i)}{kT}\right)$$

$$p_i = N_V \exp\left(\frac{-(E_i - E_v)}{kT}\right)$$

Combining the formulae:

$$\left.\begin{aligned} n &= N_C \exp\left(\frac{-(E_c - E_F)}{kT}\right) \\ n_i &= N_C \exp\left(\frac{-(E_c - E_i)}{kT}\right) \end{aligned}\right\} \rightarrow \frac{n}{n_i} = \exp\left(\frac{-(E_c - E_F) + (E_c - E_i)}{kT}\right) = \exp\left(\frac{(E_F - E_i)}{kT}\right)$$

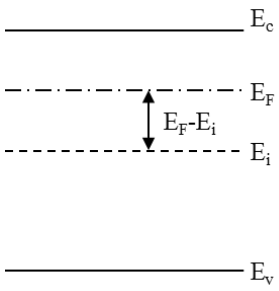
$$n = n_i \exp\left(\frac{(E_F - E_i)}{kT}\right)$$

$$\left.\begin{aligned} p &= N_V \exp\left(\frac{-(E_F - E_v)}{kT}\right) \\ p_i &= N_V \exp\left(\frac{-(E_i - E_v)}{kT}\right) \end{aligned}\right\} \rightarrow \frac{p}{p_i} = \exp\left(\frac{(E_v - E_F) - (E_v - E_i)}{kT}\right) = \exp\left(\frac{(E_i - E_F)}{kT}\right)$$

$$p = p_i \exp\left(-\frac{(E_F - E_i)}{kT}\right)$$

14. Sketch the energy band diagram for n-type Si and include E_i.

For n-Si:



15. Derive the position of E_i as a function of the bandgap.

The definition of E_i is that it is the position of E_F for intrinsic Si. For intrinsic Si we know that the carrier concentration of electrons and holes is equal. Thus we can write:
n = p
From the formulae list:

$$n = N_C \exp\left(\frac{-(E_c - E_F)}{kT}\right)$$

$$p = N_V \exp\left(\frac{-(E_F - E_v)}{kT}\right)$$

Thus:

$$n_i = N_C \exp\left(\frac{-(E_c - E_i)}{kT}\right)$$

$$p_i = N_V \exp\left(\frac{-(E_i - E_v)}{kT}\right)$$

$$n_i = p_i$$

$$N_C \exp\left(\frac{-(E_c - E_i)}{kT}\right) = N_V \exp\left(\frac{-(E_i - E_v)}{kT}\right)$$

$$\frac{N_C}{N_V} = \exp\left(\frac{-(E_i - E_v)}{kT} - \frac{-(E_c - E_i)}{kT}\right) = \exp\left(\frac{-E_i + E_v + E_c - E_i}{kT}\right)$$

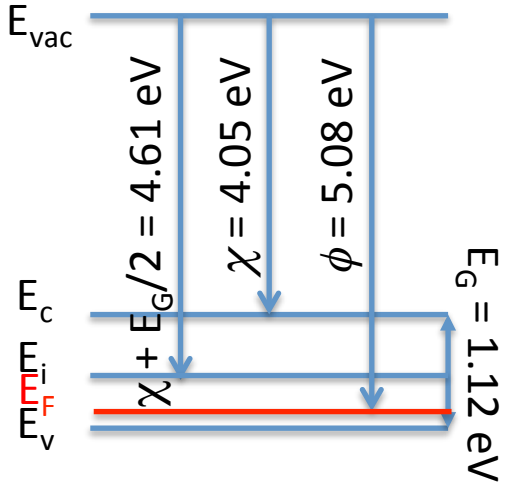
Taking E_v=0 (reference level) then E_c becomes E_G

$$\frac{N_V}{N_C} = \exp\left(\frac{2E_i + E_G}{kT}\right)$$

$$\frac{2E_i - E_G}{kT} = \ln\left(\frac{N_V}{N_c}\right)$$

$$E_i = \frac{E_G}{2} + \frac{kT}{2}\ln\left(\frac{N_V}{N_c}\right)$$

16. Calculate the value of the doping density if the workfunction of the Si sample is $\phi = 5.08$ eV. The electron affinity of Si is: $\chi = 4.05$ eV.
Start by sketching an energy band diagram: first E_{vac}, followed by E_c via knowledge of χ. Then draw E_v via knowledge of E_G. Draw the intrinsic level E_i (midgap), now draw ϕ.



This energy band diagram shows that we are working with p-Si.
The relationship between doping density and carrier concentration can be found in the formulae sheet:

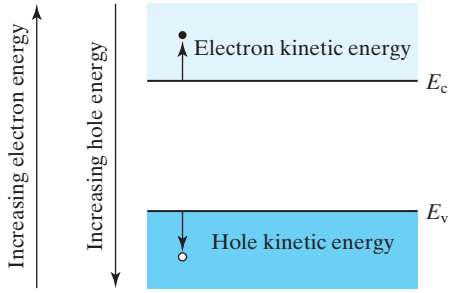
$$p = N_v e^{\frac{(E_v - E_F)}{kT}}$$

from energy band diagram: $-(E_v - E_F) = (E_{vac} - E_v) + (E_{vac} - E_F) = (\chi + E_G) - \phi = 4.05 + 1.12 - 5.08 = 0.09$ eV

$p = 1.8\times10^{19} \times \exp(-0.09/0.026) = 5.65\times10^{17}$ cm^{-3}

Thus, doping density is: $N_A = 5.65\times10^{17}$ cm^{-3}

17. Electrons with higher energies are sitting higher up in the conduction band. Holes with higher energy are sitting deeper into the valence band.



18. Fig. 5 shows a homogeneously doped p-Si material with an external voltage applied. Assuming that the contacts between the wiring and the semiconductor do not have a voltage drop, plot the energy band diagram of this Si material.

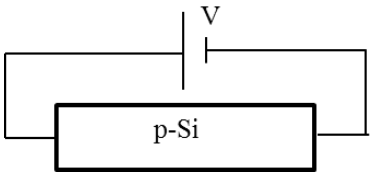
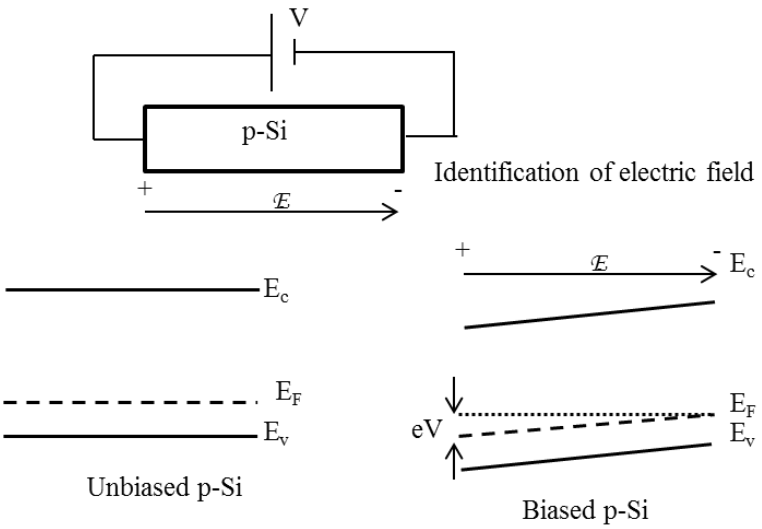


Figure 5: p-Si with biasing circuit.



Note that the holes will flow from left to right.

As a general rule, we will not plot the Fermi level in those regions where an electric field exists as this requires the introduction of the principle of quasi Fermi levels.